Experiment No. : 2

Experiment Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Experiment Program Code

Code 1: using_datatype.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script>
        let str = "Hello";
        let num = 42;
        let isReady = true;
        let empty = null;
        let notDefined;

        console.log(typeof str); // string
        document.write(str);
        console.log(typeof num); // number
        console.log(typeof isReady); // boolean
        console.log(typeof empty); // object
        console.log(typeof notDefined); // undefined
    </script>
</body>
</html>
```

Code 2: using_input_type.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2>Enter values</h2>
    x: <input type = "number" id = "num1"><br><br>
    y: <input type = "number" id = "num2"><br><br>
    z: <input type = "number" id = "num3"><br><br>

    <button onclick="showvalues()">Show Values</button>
    <p id = "result"></p>

    <script>
        function showvalues(){
            var x = Number(document.getElementById("num1").value);
            let y = Number(document.getElementById("num2").value);
            const z = Number(document.getElementById("num3").value);
            document.getElementById("result").innerHTML = "x: " + x  + "<br>" + "y: " + y + "<br>" +"z: " +
z;
```

```
            x = x + 10;
            y = y + 10;
            //z = z + 10;

            document.getElementById("result").innerText = "Updated x: " + x  + "<br>" + "Updated y: " + y +
"<br>"+ "z(const): " + z;
        }
    </script>
</body>
</html>
```

Code 3: temp.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    name: <input type = "name" id = "n1"><br><br>
    <button onclick = "showMsg">CLick This !!</button>
    <script>
        function showMsg(){
           document.getElementById("n1").innerHTML = " Welcome to SIT Nagpur" + name;
        }
    </script>
</body>
</html>
```

Code 4: destri.html

```
#Destruction of variables : array

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <script>
        const colors = ["Red ", " Green ", "Blue"];
        const [first, second] = colors;

        console.log(first);
        console.log(second);

        document.write(first );

        document.write(second);

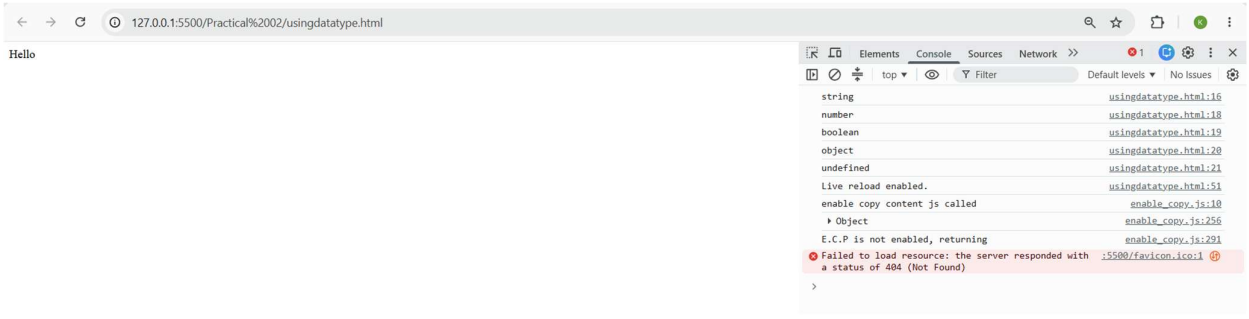
        const student = {id: 1, name: "Riya ", branch : " CSE"};
        const{name, branch} = student;

        console.log(name);
        console.log(branch);

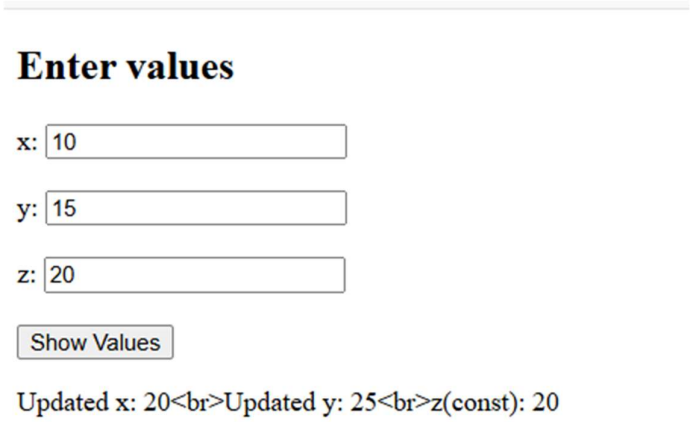
        document.write(name );
        document.write(branch);
    </script>
</body>
</html>
```

4. Output

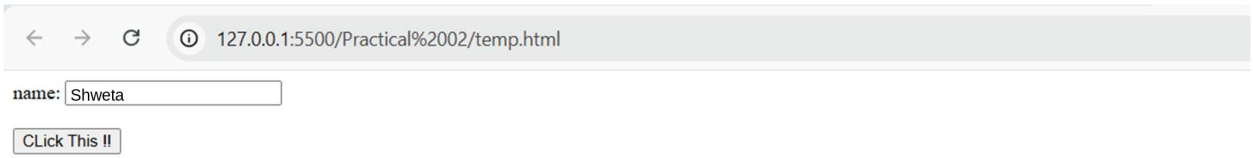
Code 1: using_datatype.html



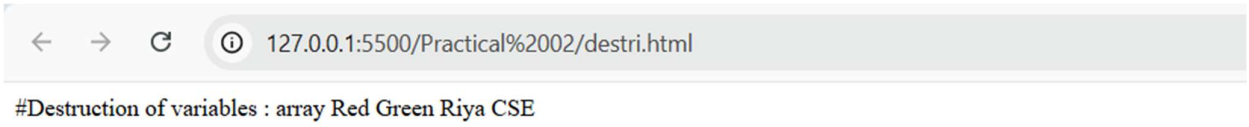
Code 2: InBodySection.html (Internal JS in Head Section)



Code 3: Windows1.html (window alert())



Code 4: WindowsPrint.html (window.print())



5. Case Study Title

Online Shopping website with billing system

6. Case Study Program Code

Casestudy_02_Shweta_Kushwaha_1001.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>PRN: 24070521001 | Name: Shweta Kushwaha | Case Study 02 — Terra &amp;amp; Leaf</title>
<meta name="description" content="A field-guide nursery. Browse and buy healthy indoor plants, delivered
pan-India.">
<link rel="preconnect" href="https://fonts.googleapis.com">
<link
href="https://fonts.googleapis.com/css2?family=Fraunces:ital,opsz,wght@0,9..144,400;0,9..144,600;1,9..144,50
0&family=Inter:wght@400;500;600;700&family=IBM+Plex+Mono:wght@400;500&display=swap" rel="stylesheet">
<link rel="stylesheet" href="styles.css">
</head>
<body>

<!-- ===== HEADER ===== -->
<header class="site-header" id="siteHeader">
  <div class="wrap header-inner">
    <a href="#top" class="logo">Terra <span>&amp;amp;</span> Leaf</a>

    <nav class="main-nav" id="mainNav" aria-label="Primary">
      <a href="#shop">Shop</a>
      <a href="#guide">Care Guide</a>
      <a href="#about">About</a>
      <a href="#journal">Journal</a>
    </nav>

    <div class="header-actions">
      <button class="icon-btn" id="cartToggle" aria-label="Open cart" aria-haspopup="dialog">
        <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.6"><path d="M4 7h16l-1.5 11a2 2 0 0 1-2 1.8H7.5a2 2 0 0 1-2-1.8L4 7Z"/><path d="M8 7V5a4 4 0 0 1 8
0v2"/></svg>
        <span class="cart-count" id="cartCount">0</span>
      </button>
      <button class="icon-btn nav-toggle" id="navToggle" aria-label="Open menu" aria-expanded="false">
        <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.6"><path d="M3 6h18M3 12h18M3 18h18"/></svg>
      </button>
    </div>
  </div>
</header>

<main id="top">

  <!-- ===== HERO ===== -->
  <section class="hero">
    <div class="wrap hero-inner">
      <div class="hero-copy">
        <p class="eyebrow">Field notes for indoor growers &mdash; No. 001</p>
        <h1>Bring the<br>outside <em>in.</em></h1>
```

```
        <p class="hero-sub">A small, honest catalog of houseplants &mdash; hand-picked, quarantine-checked,
and shipped with a printed care card. Currently stocking 42 species.</p>
        <div class="hero-cta">
          <a href="#shop" class="btn btn-primary">Browse specimens</a>
          <a href="#guide" class="btn btn-ghost">Read the care guide</a>
        </div>
      </div>
      <figure class="hero-art">
        <img src="images/monstera.jpg" alt="A Monstera deliciosa plant in a terracotta pot against a plain
wall">
        <figcaption>Monstera deliciosa <span>— specimen No. 014</span></figcaption>
      </figure>
    </div>
    <svg class="vein-divider" viewBox="0 0 1200 40" preserveAspectRatio="none" aria-hidden="true">
      <path d="M0 20 C 200 5, 300 35, 500 20 S 800 5, 1000 20 S 1150 30, 1200 18" stroke="currentColor"
fill="none" stroke-width="1"/>
    </svg>
  </section>

  <!-- ===== TRUST STRIP ===== -->
  <section class="trust-strip">
    <div class="wrap trust-grid">
      <div class="trust-item">
        <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.5"><path d="M3 7h13l4 4v6h-2M3 7v10h2M3 7l2-3h9l2 3"/><circle cx="7.5" cy="17.5" r="1.5"/><circle
cx="17.5" cy="17.5" r="1.5"/></svg>
        <p>Free delivery over ₹999</p>
      </div>
      <div class="trust-item">
        <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.5"><path d="M12 2v6M12 22v-6M4.9 4.9l4.2 4.2M14.9 14.9l4.2 4.2M2 12h6M16 12h6M4.9 19.1l4.2-4.2M14.9
9.1l4.2-4.2"/></svg>
        <p>7-day health guarantee</p>
      </div>
      <div class="trust-item">
        <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.5"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-
15A2.5 2.5 0 0 1 6.5 2Z"/></svg>
        <p>Printed care guide with every order</p>
      </div>
      <div class="trust-item">
        <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.5"><path d="M12 21s-7-4.35-9.5-8.5C.7 8.6 2.8 5 6.5 5c2 0 3.3 1 4 2.2C11.2 6 12.5 5 14.5 5 18.2 5
20.3 8.6 18.5 12.5 16 16.65 12 21 12 21Z"/></svg>
        <p>Packed by growers, not warehouses</p>
      </div>
    </div>
  </section>

  <!-- ===== SHOP ===== -->
  <section class="shop" id="shop">
    <div class="wrap">
      <div class="section-head">
        <p class="eyebrow">The catalog</p>
        <h2>In stock this week</h2>
      </div>

      <div class="filter-row" id="filterRow" role="tablist" aria-label="Filter plants">
        <button class="filter-btn is-active" data-filter="all" role="tab" aria-selected="true">All</button>
        <button class="filter-btn" data-filter="low-light" role="tab" aria-selected="false">Low
light</button>
        <button class="filter-btn" data-filter="air-purifying" role="tab" aria-selected="false">Air
purifying</button>
        <button class="filter-btn" data-filter="pet-safe" role="tab" aria-selected="false">Pet safe</button>
        <button class="filter-btn" data-filter="succulent" role="tab" aria-
selected="false">Succulents</button>
      </div>

      <div class="plant-grid" id="plantGrid"><!-- cards injected by script.js --></div>
    </div>
  </section>

  <!-- ===== CARE GUIDE CALLOUT ===== -->
  <section class="care-callout" id="guide">
    <svg class="vein-divider vein-divider--flip" viewBox="0 0 1200 40" preserveAspectRatio="none" aria-
hidden="true">
```

```
      <path d="M0 20 C 200 5, 300 35, 500 20 S 800 5, 1000 20 S 1150 30, 1200 18" stroke="currentColor"
fill="none" stroke-width="1"/>
    </svg>
    <div class="wrap care-inner">
      <div class="care-tag">Field note</div>
      <h2>Reading the tags</h2>
      <p>Every specimen card carries three marks: a sun for light needs, a drop for watering frequency, and
a leaf for growth pace. No jargon, no guesswork &mdash; just what the plant asked for.</p>
      <div class="care-legend">
        <div class="legend-item">
          <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.5"><circle cx="12" cy="12" r="4"/><path d="M12 2v3M12 19v3M4.2 4.2l2.1 2.1M17.7 17.7l2.1 2.1M2
12h3M19 12h3M4.2 19.8l2.1-2.1M17.7 6.3l2.1-2.1"/></svg>
          <div><strong>Light</strong><span>filled = bright, half = indirect, empty = low</span></div>
        </div>
        <div class="legend-item">
          <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.5"><path d="M12 2s6 7 6 11.5a6 6 0 0 1-12 0C6 9 12 2 12 2Z"/></svg>
          <div><strong>Water</strong><span>1&ndash;3 drops = weekly to monthly</span></div>
        </div>
        <div class="legend-item">
          <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.5"><path d="M12 22c0-6 4-8 4-14a4 4 0 0 0-8 0c0 6 4 8 4 14Z"/></svg>
          <div><strong>Growth</strong><span>how fast it fills a shelf</span></div>
        </div>
      </div>
    </div>
  </section>

  <!-- ===== JOURNAL / TESTIMONIAL ===== -->
  <section class="journal" id="journal">
    <div class="wrap journal-inner">
      <article class="journal-note">
        <span class="journal-no">No. 07</span>
        <p>&ldquo;Ordered a pothos in March, it's now trailing across my whole window. The care card
actually told me when to stop worrying.&rdquo;</p>
        <footer>&mdash; Ritika, Pune</footer>
      </article>
      <article class="journal-note">
        <span class="journal-no">No. 08</span>
        <p>&ldquo;First plant shop that told me which ones my cat could sit next to. Snake plant survived my
black thumb for a year.&rdquo;</p>
        <footer>&mdash; Devansh, Nagpur</footer>
      </article>
      <article class="journal-note">
        <span class="journal-no">No. 09</span>
        <p>&ldquo;Packaging alone deserves a shelf. Arrived upright, soil intact, not a single bent
leaf.&rdquo;</p>
        <footer>&mdash; Aisha, Bengaluru</footer>
      </article>
    </div>
  </section>

  <!-- ===== NEWSLETTER ===== -->
  <section class="newsletter">
    <div class="wrap newsletter-inner">
      <div>
        <p class="eyebrow">Stay rooted</p>
        <h2>Seasonal notes, once a month.</h2>
        <p class="hero-sub">Repotting reminders, new arrivals, nothing else.</p>
      </div>
      <form class="newsletter-form" id="newsletterForm">
        <label for="newsletterEmail" class="sr-only">Email address</label>
        <input type="email" id="newsletterEmail" placeholder="you@email.com" required>
        <button type="submit" class="btn btn-primary">Subscribe</button>
      </form>
      <p class="form-msg" id="newsletterMsg" role="status"></p>
    </div>
  </section>

</main>

<!-- ===== FOOTER ===== -->
<footer class="site-footer" id="about">
  <div class="wrap footer-inner">
    <div class="footer-brand">
      <a href="#top" class="logo logo--light">Terra <span>&amp;</span> Leaf</a>
```

```
      <p>A small nursery catalog, packed and shipped from Nagpur. Est. 2026.</p>
    </div>
    <div class="footer-col">
      <h3>Shop</h3>
      <a href="#shop">All plants</a>
      <a href="#shop">Low light</a>
      <a href="#shop">Pet safe</a>
    </div>
    <div class="footer-col">
      <h3>Company</h3>
      <a href="#guide">Care guide</a>
      <a href="#journal">Journal</a>
      <a href="#about">About</a>
    </div>
    <div class="footer-col">
      <h3>Contact</h3>
      <a href="mailto:hello@terraandleaf.example">hello@terraandleaf.example</a>
      <a href="tel:+910000000000">+91 00000 00000</a>
    </div>
  </div>
  <div class="wrap footer-bottom">
    <p>&copy; 2026 Terra &amp; Leaf. All plants, no filler.</p>
  </div>
</footer>

<!-- ===== CART DRAWER ===== -->
<div class="cart-backdrop" id="cartBackdrop"></div>
<aside class="cart-drawer" id="cartDrawer" role="dialog" aria-label="Shopping cart" aria-hidden="true">
  <div class="cart-drawer-head">
    <h2>Your cart</h2>
    <button class="icon-btn" id="cartClose" aria-label="Close cart">
      <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.6"><path d="M4 4l16 16M20 4L4 20"/></svg>
    </button>
  </div>
  <div class="cart-items" id="cartItems">
    <p class="cart-empty" id="cartEmpty">Your cart is empty. Go find a plant.</p>
  </div>
  <div class="cart-footer">
    <div class="cart-subtotal">
      <span>Subtotal</span>
      <span id="cartSubtotal">₹0</span>
    </div>
    <button class="btn btn-primary btn-block" id="checkoutBtn">Checkout</button>
  </div>
</aside>

<!-- ===== INVOICE / CHECKOUT PANEL ===== -->
<div class="invoice-backdrop" id="invoiceBackdrop"></div>
<section class="invoice-panel" id="invoicePanel" role="dialog" aria-label="Invoice and billing" aria-
hidden="true">
  <div class="invoice-panel-head">
    <h2>Invoice &amp; Billing</h2>
    <button class="icon-btn" id="invoiceClose" aria-label="Close invoice">
      <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="1.6"><path d="M4 4l16 16M20 4L4 20"/></svg>
    </button>
  </div>

  <div class="invoice-panel-body">
    <form class="invoice-form" id="invoiceForm" onsubmit="return false;">
      <fieldset class="invoice-fieldset">
        <legend>Customer details</legend>
        <table class="form-table">
          <tr>
            <td><label for="custName">Customer name</label></td>
            <td><input type="text" id="custName" name="custName" required placeholder="Full name"></td>
          </tr>
          <tr>
            <td><label for="mobile">Mobile no</label></td>
            <td><input type="tel" id="mobile" name="mobile" required placeholder="10-digit number"
pattern="[0-9]{10}" maxlength="10"></td>
          </tr>
          <tr>
            <td><label for="invoiceNo">Invoice no</label></td>
            <td><input type="text" id="invoiceNo" name="invoiceNo" readonly></td>
          </tr>
```

```
          <tr>
            <td><label for="invoiceDate">Invoice date</label></td>
            <td><input type="date" id="invoiceDate" name="invoiceDate" required></td>
          </tr>
        </table>
      </fieldset>

      <fieldset class="invoice-fieldset">
        <legend>Products (from cart)</legend>
        <div class="invoice-table-wrap">
          <table class="invoice-products" id="invoiceProducts">
            <thead>
              <tr>
                <th>Product name</th>
                <th>Qty (units)</th>
                <th>Rate (₹)</th>
                <th>Line total (₹)</th>
              </tr>
            </thead>
            <tbody id="invoiceProductBody">
              <!-- rows filled by script.js -->
            </tbody>
          </table>
        </div>
      </fieldset>

      <fieldset class="invoice-fieldset">
        <legend>Billing options</legend>
        <table class="form-table">
          <tr>
            <td><label for="discount">Discount %</label></td>
            <td><input type="number" id="discount" name="discount" min="0" max="100" value="0"></td>
          </tr>
          <tr>
            <td><label for="gst">GST %</label></td>
            <td><input type="number" id="gst" name="gst" min="0" max="28" value="18"></td>
          </tr>
          <tr>
            <td><label for="packing">Packing charges (₹)</label></td>
            <td><input type="number" id="packing" name="packing" min="0" value="50"></td>
          </tr>
          <tr>
            <td>Payment mode</td>
            <td class="radio-group">
              <label><input type="radio" name="paymentMode" value="Cash" checked> Cash</label>
              <label><input type="radio" name="paymentMode" value="Card"> Card</label>
              <label><input type="radio" name="paymentMode" value="UPI"> UPI</label>
            </td>
          </tr>
          <tr>
            <td>Membership</td>
            <td class="radio-group">
              <label><input type="radio" name="membership" value="no" checked> No</label>
              <label><input type="radio" name="membership" value="yes"> Yes</label>
            </td>
          </tr>
        </table>
      </fieldset>

      <div class="invoice-actions">
        <button type="button" class="btn btn-primary" id="calcBillBtn" onclick="calculateBill()">Calculate
Bill</button>
        <button type="button" class="btn btn-ghost" id="printBillBtn" onclick="printInvoice()">Print
Invoice</button>
        <button type="reset" class="btn btn-ghost">Reset form</button>
      </div>
    </form>

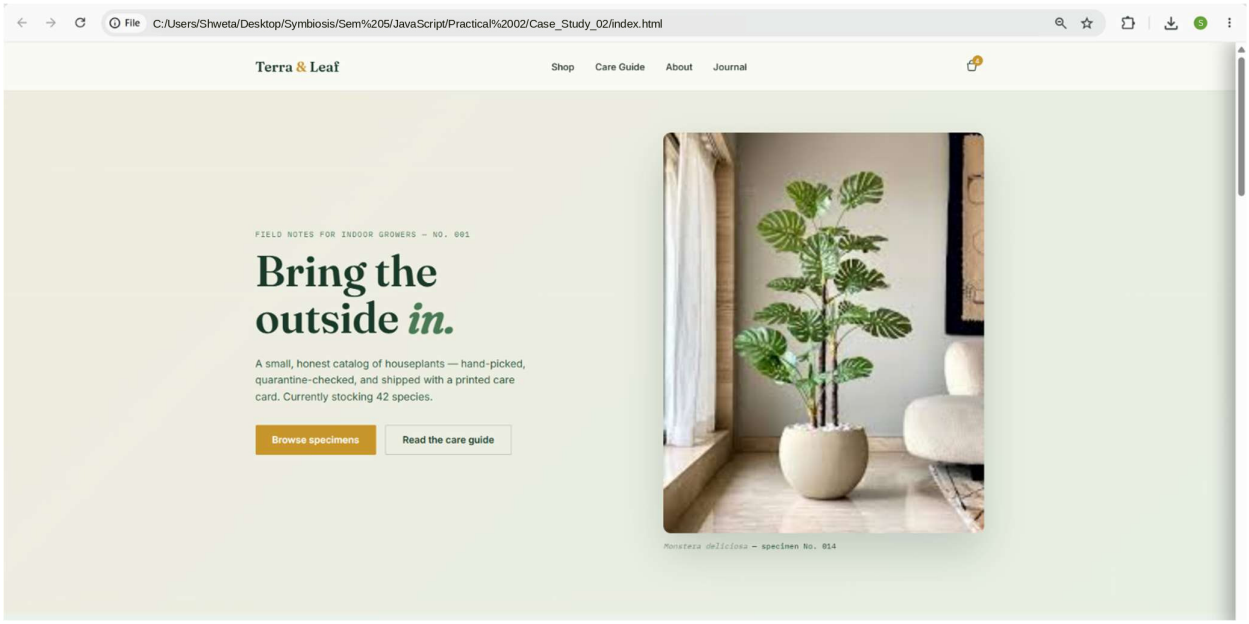
    <section class="bill-receipt" id="billReceipt" aria-live="polite">
      <p class="bill-receipt-placeholder">Fill the form and click <strong>Calculate Bill</strong> to see
your receipt here.</p>
    </section>
  </div>
</section>

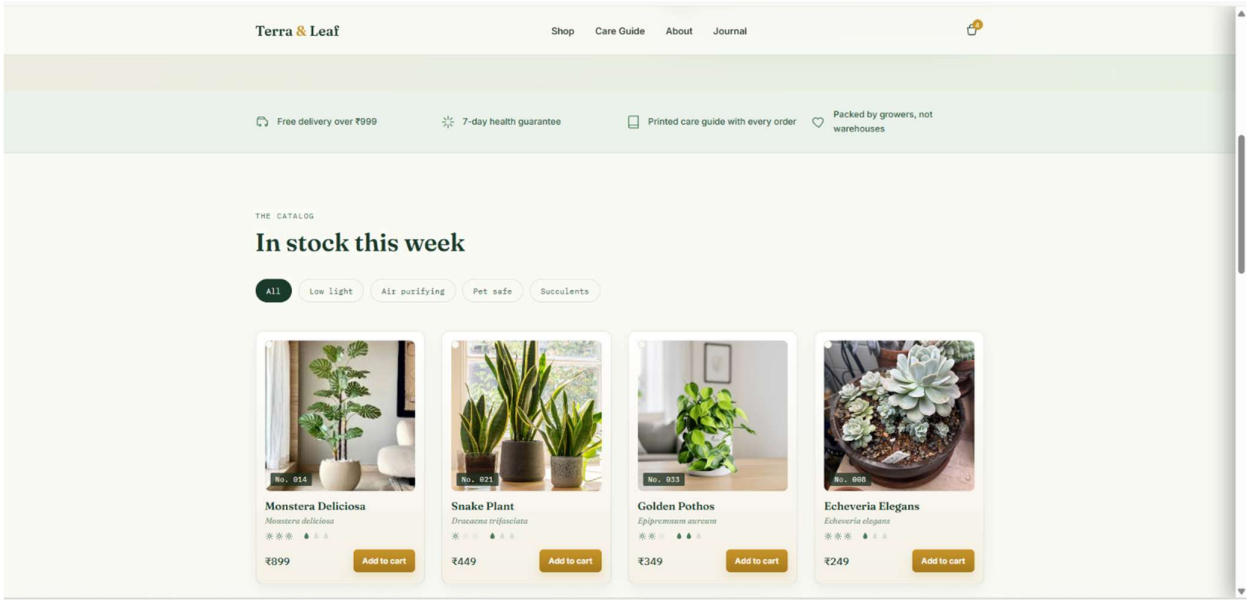
<script src="script.js"></script>
</body>
```

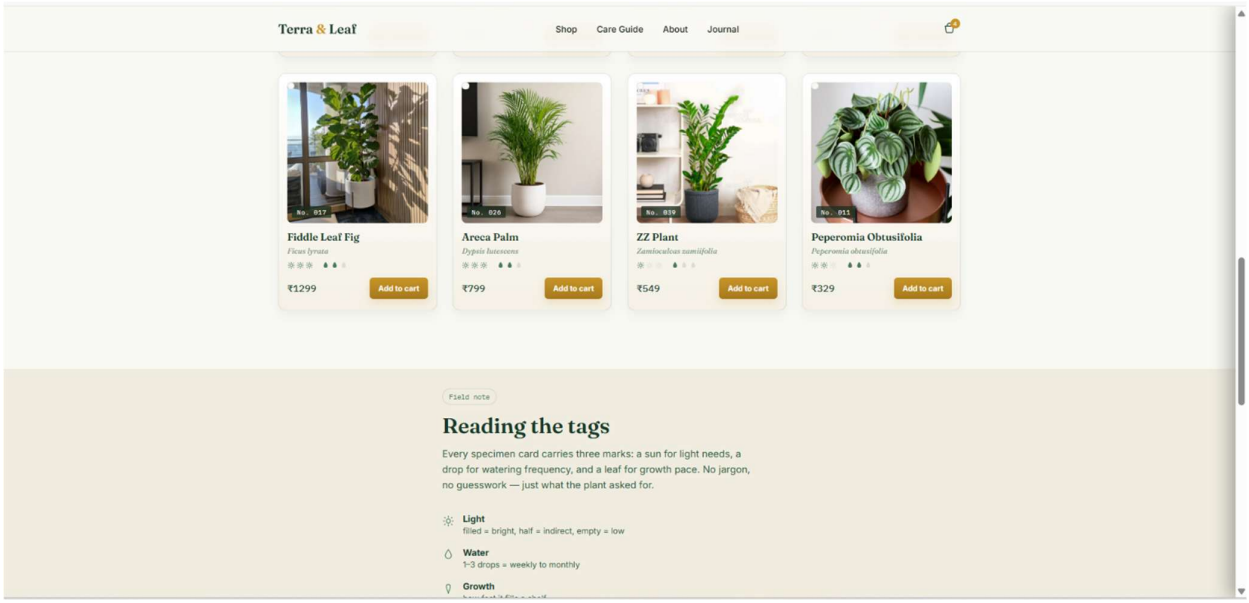
```
</html>
```

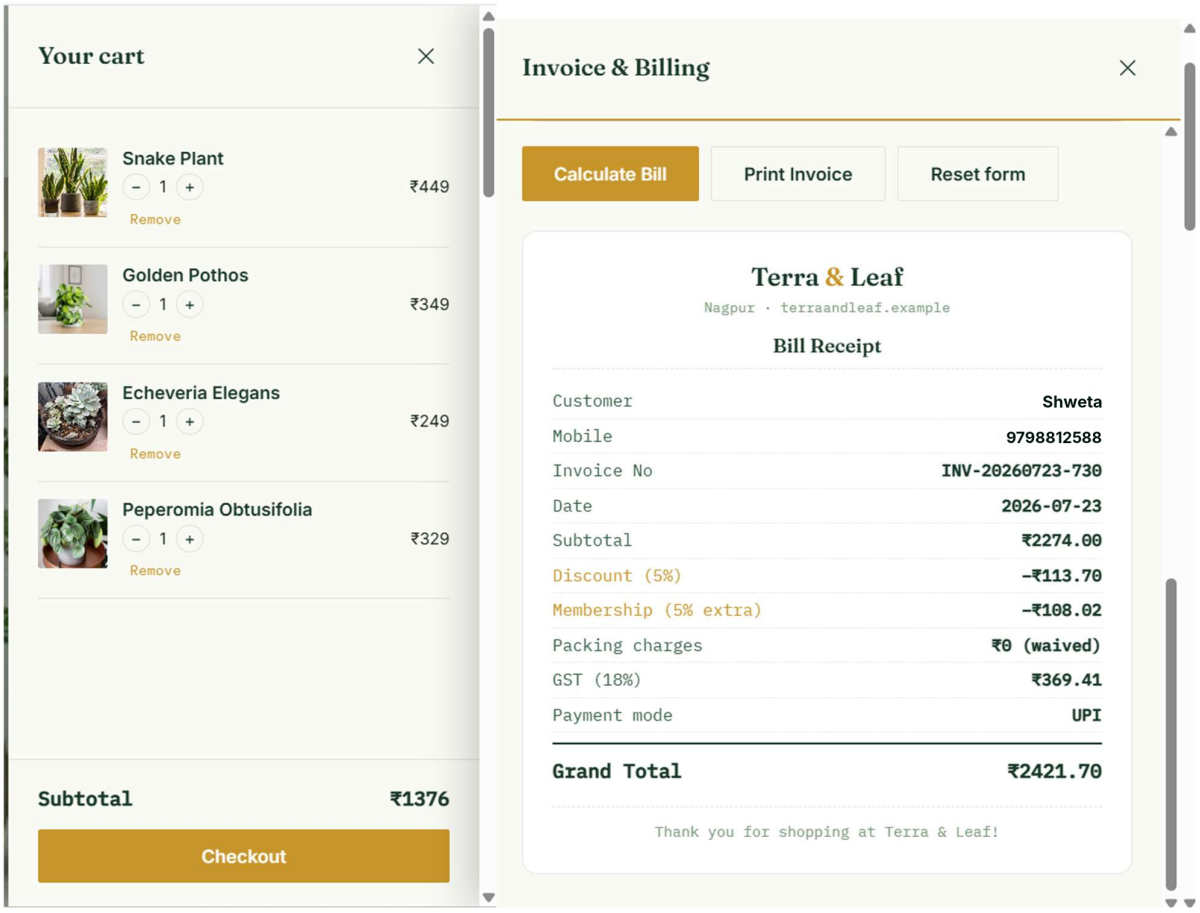
7. Output of the case study :

Name : Shweta Kushwaha PRN : 24070521001 File path :
http://127.0.0.1:5500/Practical%2002/Casestudy_02_Shweta_Kushwaha_1001.html









8. Result/Conclusion

Thus, the differences between var, let, and const were demonstrated along with their scope behavior. Template literals (string interpolation, multi-line strings) and destructuring (array and object) were also implemented successfully